

NCA, LFP, and graphite, as well as silicon-graphite blended anodes. The prismatic form factor, with its rigid metal casing, is preferred for electric mobility applications because it provides mechanical stability and allows for fewer units per battery pack, simplifying diagnostics and assembly at the pack level and opening the doors to advanced edge compute-based AI applications such as IBC's proprietary battery passport program.

While discussing the design challenges, Priyadarshi adds, "When building our solution, it was critical to understand its purpose, and that's where design challenges emerged. Our target yields were to achieve industry-leading (above 90%) yields, but due to proprietary designs and excellence in manufacturing, we are now achieving 94-95% end of line yields in South Korea. Designing with scalability in mind required careful trade-offs. To navigate this, we prioritised customer pain points using voice-of-customer analysis and scorecards, accepting compromises only on less critical criteria. Through this rigorous approach, we addressed design trade-offs and challenges, ultimately creating a quality product that can be manufactured reliably at scale, and the technology can be transferred seamlessly to mega factories using our copy-exact scaling methodology."

Currently, the company's high-volume manufacturing takes place at its South Korea facility, where the battery cells are produced.



Priyadarshi Panda, Founder and CEO, International Battery Company

These cells are then shipped to India, where the company assembles two-wheeler battery packs for local deployment through a proof of concept pack design shareable with pack and cell partners. Through its partnership with Mahanagar Gas Limited, the company is implementing an "end-to-end copy exact" approach, transferring the technology, processes, and specifications from Korea to India at scale.

Regarding the revenue, Priyadarshi reveals, "In the 2025 calendar year, we deployed over 1000 two-wheeler battery packs in the field. The assets currently in use are valued at nearly ₹60 million, while our order book has crossed approximately ₹200 million."

The startup highlighted certain areas where the support ecosystem can help accelerate the product technology build and growth trajectory, such as more government funding

designed for capital-heavy hardware startups focused on bringing cutting-edge product technology to the market in areas of national importance, and accelerated and preferential customer adoption in India. It would truly benefit Indian companies building Li-ion cells in India if Indian customers are strongly encouraged to choose Indian suppliers as opposed to preferring low-cost, low-quality suppliers from China. The startup claims to work closely with universities, especially in the US, where a board member is leading a project on battery-grade graphite, an important battery material. In India, such partnerships have not yet been established, but the company values working with academia for its knowledge and expertise. Strengthening ties with universities remains a key priority as the company grows.

Looking ahead, the company is working on many areas, focusing on building technology and knowledge through its innovation engine in Silicon Valley, USA. In India, it tests products designed for Indian conditions in tough, real-world conditions, like heat and rough roads, to ensure they work well. The company cares about customer success and is building partnerships with vehicle makers, customers, and local material suppliers to make products locally and at scale. It is also investing in talent, testing, partnerships, and sales, and is now expanding these efforts globally.

—Nidhi Agarwal

2 BOTLAB DYNAMICS ENGINEERING INDIA'S DRONE SWARM BREAKTHROUGH

An IIT Delhi-born startup redefining large-scale drone swarms through indigenous technology and global aerial spectacles.

Founded in 2016, New Delhi-based BotLab Dynamics Private Limited is an Indian deep-tech startup incubated at IIT

Delhi, established by Anuj Kumar Barnwal, Dr Sarita Ahlawat, and Tanmay Bunkar. The company designs and operates large-scale

autonomous drone swarms for coordinated aerial displays at major national and public events, while also developing in-house drone